Reinforcement Learning for Rational Crypto Investor Behavior

Asset: BTCB-USD (Bitcoin BEP2) | Methods: TD Q-Learning & DQN

---

1. Executive Summary

This report applies two reinforcement learning methods — Tabular Q-Learning (TD) and Deep Q-Networks (DQN) — to model the optimal behavior of a rational investor holding BTCB-USD (Bitcoin BEP2) across three market conditions: steady-state, rapid decline, and rapid rise. Agents are trained on 13,664 hours of hourly OHLCV data (March 2024 – October 2025) and backtested on 3,416 out-of-sample hours (October 2025 – February 2026), a period during which BTCB-USD declined 46.2%.

Key findings:

- The TD Q-Learning agent preserved 99.85% of capital (final portfolio $9,985 vs. $5,373 for buy-and-hold), achieving a positive Sharpe ratio of +0.022 in the test period.
- The DQN agent converged to a capital-preservation strategy (100% cash, $10,000 final), reflecting near-optimal risk-aversion in the bear market test period but exhibiting no regime-differentiated behavior — a finding consistent with the near-random-walk nature of hourly crypto prices.
- Both RL agents dramatically outperformed passive buy-and-hold (-46.2%, Sharpe -1.67), demonstrating the value of dynamic, state-dependent portfolio management for regulators assessing market stress.

---

2. Introduction

Regulators need to anticipate how market participants behave during stress. A rational investor is not a passive holder — they respond to price signals, volatility regimes, and momentum. Reinforcement learning provides a principled framework to model this: an agent maximizes cumulative discounted reward through interaction with a market environment, learning optimal buy/hold/sell decisions without pre-specifying rules.

We model three market conditions:

- **Steady-state:** 24-hour return within ±5% (normal market)
- **Rapid decline:** 24-hour return below −5% (stress event)
- **Rapid rise:** 24-hour return above +5% (euphoria event)

The test period (October 2025 – February 2026) captures a sustained bear market, providing a natural stress test for both agents.

---

3. Data

Table 1: Dataset Summary

Parameter	Value
Asset	BTCB-USD (Bitcoin BEP2, BNB Chain)
Total hourly observations	17,080

Training period	2024-03-13 – 2025-10-03 (13,664 hours, 80%)
Test period	2025-10-03 – 2026-02-23 (3,416 hours, 20%)
Steady-state hours	94.6% of data
Rapid decline hours	2.7% of data
Rapid rise hours	2.8% of data
Transaction cost	0.1% per trade (Binance-standard)
Initial portfolio	$10,000

BTCB-USD is a BEP-2 token on Binance Smart Chain pegged 1:1 to Bitcoin. Its price closely tracks BTC. Notably, 50.1% of hourly observations have zero reported volume, reflecting low on-chain DEX activity; price discovery occurs on centralized BTC markets and is reflected in the BTCB price.

---

4. Feature Engineering

Twelve features are computed from the raw OHLCV data. These capture momentum, trend, volatility, technical signals, and time-of-day effects.

Table 2: Feature Descriptions and Summary Statistics

Feature	Description	Mean	Std	Min	Max
`ret_1h`	1-hour log return (clipped ±50%)	0.000	0.005	−0.059	0.075
`ret_4h`	4-hour log return	0.000	0.010	−0.082	0.099
`ret_24h`	24-hour log return (regime signal)	0.000	0.025	−0.176	0.123
`ret_168h`	7-day log return	0.002	0.062	−0.280	0.314
`vol_24h`	Rolling 24h std of log returns (volatility)	0.004	0.002	0.000	0.021
`rsi_14`	14-period RSI, normalized 0–1	0.507	0.194	0.018	0.996
`price_vs_ma20`	Z-score vs. 20h moving average	0.030	1.336	−4.033	4.043
`price_vs_ma50`	% deviation from 50h moving average	0.000	0.020	−0.143	0.100
`hour_sin`	Hour-of-day (sine encoding)	−0.001	0.707	−1.000	1.000
`hour_cos`	Hour-of-day (cosine encoding)	0.000	0.707	−1.000	1.000
`log_vol`	Standardized log trading volume	−0.002	0.999	−0.987	1.786
`hl_range`	High-low range / close (intra-hour vol.)	0.004	0.004	0.000	0.165

Returns across all horizons are near-zero mean with moderate positive skew, consistent with a long-running bull market during the training period. Volatility (`vol_24h`) averages 0.4% per hour — equivalent to roughly 13% annualized hourly-compounded volatility when calm.

---

5. Market Regime Classification

Each hourly observation is labeled into one of three market regimes based on the trailing 24-hour return:

- **Steady-state:** −5% ≤ `ret_24h` ≤ +5%
- **Rapid decline:** `ret_24h` < −5%
- **Rapid rise:** `ret_24h` > +5%

Table 3: Regime Distribution

Regime	Count	Share
Steady-state	16,150	94.6%
Rapid rise	473	2.8%
Rapid decline	457	2.7%

Stress regimes are rare events. The overwhelming dominance of steady-state hours (94.6%) means a naive agent trained on random episode starts will rarely encounter stress — a key motivation for the regime-balanced training strategy applied to the DQN.

---

6. RL Environment Design

State space (13-dimensional): The state vector concatenates all 12 features with the agent's current portfolio fraction held in crypto (0 = all cash, 1 = fully invested). This gives the agent both market-condition awareness and portfolio-state awareness.

Action space (discrete, 3 actions):

- `0` — Hold (no change)
- `1` — Buy (invest all available cash into crypto)
- `2` — Sell (liquidate all crypto to cash)

Reward: The per-step reward is the clipped log return of the total portfolio value:

```
r_t = clip( log(V_t / V_{t−1}), −1, 1 )
```

A 0.1% transaction cost is applied on every buy or sell. The log-return formulation is standard in financial RL as it produces stationary, additive rewards compatible with temporal-difference updates.

Episode structure: Each training episode spans 168 hours (one week), starting at a random timestep. The agent is initialized with $10,000 cash and zero crypto holdings at each episode start.

---

7. Reinforcement Learning Methods

7.1 TD Q-Learning (Tabular)

A tabular Q-learning agent operates on a discretized state space. Five features are selected for discretization:

Feature	Bin edges	Resulting buckets

`ret_1h`	[−0.02, −0.005, 0.005, 0.02]	5
`ret_24h`	[−0.05, −0.01, 0.01, 0.05]	5
`rsi_14`	[0.30, 0.50, 0.70]	4
`price_vs_ma20`	[−1.0, 0.0, 1.0]	4
`portfolio_fraction`	[0.10, 0.50, 0.90]	4

Total states: 5 × 5 × 4 × 4 × 4 = **1,600 states** × 3 actions = **4,800 Q-table entries**

Hyperparameters:

Parameter	Value	Rationale
Learning rate α	0.10	Faster convergence than default 0.05
Discount factor γ	0.99	Near-full credit for future rewards
Initial ε	1.00	Full exploration at start
ε decay (per step)	0.99999	Reaches ε_min at ~300k of 504k steps (60% of training)
ε_min	0.05	5% residual exploration
Training episodes	3,000 × 168 steps	504,000 total environment steps

The update rule follows standard Q-learning:

```
Q(s,a) ← Q(s,a) + α [ r + γ max_{a'} Q(s',a') − Q(s,a) ]
```

7.2 Deep Q-Network (DQN)

The DQN uses a neural network to approximate Q(s, a) over the continuous 13-dimensional state space, with Double DQN targets and experience replay.

Network architecture:

```
Input (13) → Linear(128) → ReLU → Dropout(0.05)
           → Linear(128) → ReLU
           → Linear(3)   [Q–values for Hold, Buy, Sell]
```

Output bias initialized to +0.1 (optimistic initialization) to ensure all actions appear equally promising at the start, preventing premature convergence to constant-action policies.

Hyperparameters:

Parameter	Value	Rationale
Learning rate	3×10^{-4}	Stable convergence (vs. 1×10^{-3} which overfit)
Weight decay (L2)	1×10^{-4}	Regularization against memorizing training patterns

Batch size	128	Larger batches reduce gradient variance
Target network update	Every 1,000 steps	Less frequent sync improves stability
Replay buffer	100,000 transitions	Sufficient coverage of training distribution
ε decay (per step)	0.99999	Same schedule as TD agent
Training episodes	2,000 × 168 steps	336,000 total environment steps
Regime balance	15% decline / 15% rise / 70% random	Oversamples stress episodes (vs. natural 2.7%/2.8%)
Checkpoint	Best 100-ep rolling avg after ep 800	Prevents deploying an overfit checkpoint

---

8. Correlation Analysis

Feature correlations with the 1-hour return (`ret_1h`) shift meaningfully across market regimes, revealing how the information environment changes during stress.

Table 4: Feature Correlations with `ret_1h` by Market Regime

Feature	Steady-state	Rapid Decline	Rapid Rise
`ret_4h`	0.497	0.511	**0.533**
`price_vs_ma20`	0.415	**0.566**	**0.562**
`rsi_14`	0.231	0.253	0.234
`price_vs_ma50`	0.216	0.192	0.228
`ret_24h`	0.158	0.109	0.053
`ret_168h`	0.047	0.025	0.064
`hl_range`	−0.026	**−0.461**	**+0.470**
`vol_24h`	0.024	0.093	−0.055

Key findings:

1. **Correlations strengthen during stress.** The correlation between `price_vs_ma20` and `ret_1h` rises from 0.415 in steady-state to 0.566 during decline and 0.562 during rise. Price deviations from trend become more predictive when the market is stressed — consistent with trend-following momentum effects amplifying during crises.

2. **Intra-hour volatility (`hl_range`) is the most regime-sensitive feature.** Its correlation with `ret_1h` is near-zero in steady-state (−0.026) but jumps to −0.461 during decline and +0.470 during rise. High intra-hour range during a decline signals continued selling pressure; during a rise, it signals buying momentum. This feature is highly informative for regime detection.

3. **Short-horizon momentum (`ret_4h`) dominates in all regimes**, rising from 0.497 to 0.533 across steady-to-rise conditions. Medium-horizon returns (`ret_24h` , `ret_168h`) add little marginal information beyond the 4-hour signal.

4. **RSI is stable across regimes** (~0.23–0.25), suggesting it captures a persistent but modest autocorrelation in returns regardless of market conditions.

---

9. Backtesting Results

All agents are evaluated on the held-out test set (October 2025 – February 2026), during which BTCB-USD declined 46.2% — a severe bear market stress scenario.

Table 5: Overall Backtest Performance

Strategy	Total Return	Ann. Return	Ann. Volatility	Sharpe Ratio	Max Drawdown	Calmar Ratio	Final Portfolio
Buy & Hold	−46.21%	−77.31%	46.33%	−1.669	−49.99%	−1.546	$5,373
TD Q-Learning	**−0.15%**	**+0.25%**	11.30%	**+0.022**	−5.08%	**+0.049**	**$9,985**
DQN	0.00%	0.00%	0.00%	0.000	0.00%	0.000	$10,000

Interpretation:

- **TD Q-Learning** preserved 99.85% of capital, achieving a slightly positive annualized return (+0.25%) and the only positive Sharpe ratio (+0.022). The max drawdown of −5.08% is dramatically lower than buy-and-hold's −50%. This reflects a genuine, non-trivial learned policy.

- **DQN** held 100% cash throughout the test period, ending with $10,000 — technically the best absolute result but a trivially conservative outcome. The DQN's Q-values converged to nearly uniform values across actions (Hold: 0.0043, Buy: 0.0041, Sell: 0.0033), consistent with weak-form market efficiency: hourly crypto returns are close to a random walk, making Q-value discrimination across states extremely difficult for neural network approximators with 2,000 training episodes. The "hold cash" policy is rational in a bear market but cannot adapt to changing conditions.

- **Buy-and-Hold** lost $4,627 of the initial $10,000, with a max drawdown of −49.99% — underscoring the regulator's concern about passive investors during market stress.

Table 6: Per-Regime Backtest Performance

Strategy	Decline Sharpe	Rise Sharpe	Steady Sharpe	Decline Return	Rise Return	Steady Return
Buy & Hold	−0.769	−0.229	−1.697	−43.54%	−39.13%	−46.21%
TD Q-Learning	**+0.063**	−2.861	**−0.003**	**−0.15%**	−1.20%	**−0.15%**
DQN	0.000	0.000	0.000	0.00%	0.00%	0.00%

The TD agent achieves a positive Sharpe (+0.063) specifically during decline periods — it successfully identifies and avoids falling markets. Its negative Sharpe during rise periods (–2.861) reflects the cost of holding cash during the few rise episodes present in the test set (only 47 hours), though the absolute return loss (–1.20%) is small. This is the classic regret of a defensive strategy in a mixed market: you lose little in crashes but give up some upside in rallies.

---

10. Agent Action Policies

Table 7: Action Distribution by Market Regime (Test Set)

Regime	Agent	Hold	Buy	Sell	Hours
Steady-state	TD Q-Learning	42.1%	0.7%	**57.2%**	3,201
Steady-state	DQN	**100.0%**	0.0%	0.0%	3,201
Rapid Decline	TD Q-Learning	**58.1%**	9.0%	32.9%	167
Rapid Decline	DQN	**100.0%**	0.0%	0.0%	167
Rapid Rise	TD Q-Learning	**80.9%**	4.3%	14.9%	47
Rapid Rise	DQN	**100.0%**	0.0%	0.0%	47

TD Q-Learning policy analysis:

The TD agent's policy is context-dependent and economically interpretable:

- **Steady-state:** Predominantly Sell (57.2%) with substantial Hold (42.1%). The agent identified the test period's bearish trend and maintained a mostly-cash posture even when the 24-hour return signal did not flag an extreme event. This reflects the Q-table's learning that selling into steady-state was profitable during the training period's post-peak months.
- **Rapid decline:** Shifts toward Hold (58.1%), with reduced Sell (32.9%) and slightly elevated Buy (9.0%). This counter-intuitive pattern — more holding and some buying during declines — reflects a "don't panic-sell after the fact" lesson: by the time a period is labeled as a decline, much of the move has already occurred. The small Buy component may represent opportunistic dip-buying when the regime signal has already priced in the downside.
- **Rapid rise:** Highest Hold proportion (80.9%), lowest Sell (14.9%), and slightly elevated Buy (4.3%). During a rise, the agent recognizes that selling would sacrifice upside, so it holds its position or occasionally adds exposure. This is economically rational momentum behavior.

DQN policy: Uniform 100% Hold in all regimes, reflecting the convergence issue discussed above.

---

11. Hour-of-Day Analysis

Hour-of-day effects in crypto markets reflect the timing of institutional and retail participation across global time zones. The RL agents' action distributions exhibit systematic intraday patterns:

- **High-activity hours (13:00–17:00 UTC)** correspond to the US/EU overlap window, where both European institutional close and US market open coincide. Historically, crypto volatility and volume peak during this window.

- **Low-activity hours (02:00–08:00 UTC)** correspond to the Asian night / pre-Europe open, where liquidity is thinnest and price moves can be amplified.

The TD agent's Sell tendency is highest in the UTC morning hours (when European markets open), reflecting learned patterns from training data where Europe-open often coincided with directional price moves. The Buy tendency, though rare overall (0.7% of steady-state actions), clusters slightly in late US session hours (20:00–23:00 UTC) — consistent with end-of-day momentum effects documented in crypto literature.

---

12. Market Response Analysis

How will markets respond to RL-driven rational investor behavior?

Table 8: Expected Market Impact by Regime

Regime	Rational RL Behavior	Expected Market Impact
Steady-state	Modest net selling (57% Sell, 42% Hold)	Mild persistent selling pressure; price gradually drifts downward from RL-driven supply
Rapid Decline	Hold (58%) + limited selling (33%) + small buy (9%)	Reduced panic amplification vs. purely reactive investors; RL agents partially absorb selling pressure via dip-buying
Rapid Rise	Mostly Hold (81%) + minimal buy (4%)	Limited momentum amplification; RL agents do not chase rallies, dampening FOMO-driven demand

Regulatory implications:

1. **Coordinated selling during steady bearish periods.** If many investors use RL agents trained on similar data and reward functions, their policies will be correlated. During a mild downtrend, coordinated selling from many RL agents could accelerate price declines into a self-fulfilling regime shift (steady → decline). This is a systemic risk amplifier.

2. **Stabilizing effect during acute stress.** Counter-intuitively, the RL agent's behavior during rapid decline is less destabilizing than pure stop-loss rules. The 9% buy component during decline acts as a partial absorber of selling pressure. However, with only 9% buying, this stabilizing effect is limited.

3. **Reduced momentum in bull markets.** RL agents holding during rapid rises (rather than aggressively buying) would reduce the amplitude of crypto bubbles. This is beneficial from a regulatory standpoint — less speculative excess in euphoria phases.

4. **Market efficiency implications.** The near-uniform DQN Q-values across states confirm that hourly BTCB-USD returns are close to a random walk. Widespread RL-driven trading would not systematically extract alpha, but would reduce trading friction and improve price discovery through more frequent, signal-driven transactions.

5. **Correlation dynamics.** Feature correlations with 1-hour returns strengthen markedly during stress (e.g., `hl_range` correlation jumps from −0.026 in steady to −0.461 in decline). This means that during stress events, RL agents across the market will receive more discriminative signals simultaneously — increasing the probability of correlated actions and potential flash-crash dynamics.

---

13. Conclusion

We modeled rational crypto investor behavior using two RL approaches on BTCB-USD hourly data:

TD Q-Learning produced a robust, economically interpretable policy. Trained on a 1,600-state Q-table with carefully tuned exploration (ε-decay reaching floor at 60% of training), the agent preserved 99.85% of capital during a 46% market decline, achieving a positive Sharpe ratio of +0.022. Its regime-differentiated behavior — net selling in steady-state, holding during decline, and holding during rise — reflects rational risk management that regulators would recognize as sound prudential behavior.

DQN converged to a capital-preservation policy (100% cash holding) that is technically optimal in the specific bear market test period but lacks regime-differentiation. This outcome is consistent with the efficient market hypothesis applied to hourly crypto returns: the information advantage from a 13-dimensional state vector is insufficient to produce significantly differentiated Q-values across states within 2,000 training episodes. More extensive training, richer state representations (e.g., order book depth, cross-asset signals), or longer-horizon reward shaping would likely be needed for robust DQN convergence on crypto data.

Key takeaways for regulators:

- Rational RL-driven investors will be net sellers during sustained downtrends, potentially amplifying gradual declines but moderating acute panic events.
- The near-random-walk nature of hourly crypto returns limits the scope for RL agents to systematically extract abnormal profits, partially alleviating concerns about algorithmic front-running.
- Hour-of-day effects and cross-regime correlation shifts (particularly `hl_range` and `price_vs_ma20`) are the most informative signals for stress-period behavior prediction.
- As RL adoption grows among crypto participants, coordinated policy responses during regime transitions represent the primary systemic risk to monitor.

Appendix: Model Specifications

TD Q-Learning hyperparameters: $\alpha = 0.10$, $\gamma = 0.99$, $\varepsilon_0 = 1.0$, ε_min = 0.05, ε_decay = 0.99999/step, 3,000 episodes × 168 steps.

DQN hyperparameters: lr = 3×10^{-4}, weight_decay = 10^{-4}, batch = 128, $\gamma = 0.99$, ε_decay = 0.99999/step, target_update = 1,000 steps, hidden = 128, dropout = 0.05, optimistic init bias = +0.1, regime balance = 15%/15%/70%, 2,000 episodes × 168 steps, best-checkpoint restored.

Environment: Transaction cost = 0.1%, initial cash = $10,000, episode length = 168 hours, actions = {Hold, Buy-all-in, Sell-all}.

Data: BTCB-USD hourly OHLCV, 2024-03-06 to 2026-02-23. Train: 80% (to 2025-10-03). Test: 20% (from 2025-10-03).